

Problem 4: Sets Whose Pairwise Sums Are Never Squarefree

Standalone statement and proof

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Abstract

We prove the conjectured asymptotic for Problem 4 from the curated list of ten AI-Erdős-style problems. In fact, we prove a slightly stronger eventual exact result: if $Q(N)$ denotes the largest size of a set $A \subseteq [N]$ such that $a + b$ is not squarefree for all distinct $a, b \in A$, then

$$Q(N) = \left\lfloor \frac{N+2}{4} \right\rfloor$$

for all sufficiently large N . The proof is elementary and uses only the standard Möbius-function identity for the squarefree indicator, the Chinese remainder theorem, and a modulo 4 classification.

1 Statement of the problem

For $N \geq 1$, write

$$[N] = \{1, 2, \dots, N\}.$$

A positive integer is called *squarefree* if no square of a prime divides it.

Problem 4: non-squarefree pair-sum sets. Call a set $A \subseteq [N]$ *non-squarefree-sum* if for every pair of distinct elements $a, b \in A$, the sum $a + b$ is not squarefree. Let

$$Q(N) = \max\{|A| : A \subseteq [N] \text{ is non-squarefree-sum}\}.$$

Is

$$Q(N) = \left(\frac{1}{4} + o(1)\right) N?$$

The elementary construction in the original problem is the residue class $0 \pmod{4}$: if

$$A = \{n \leq N : n \equiv 0 \pmod{4}\},$$

then every pair-sum is divisible by 4, and hence is not squarefree. This gives $Q(N) \geq N/4 + O(1)$. For the eventual exact statement it is slightly better, depending on $N \pmod{4}$, to use the residue class $2 \pmod{4}$.

2 Main result

Theorem 1. *For all sufficiently large N ,*

$$Q(N) = \left\lfloor \frac{N+2}{4} \right\rfloor.$$

Consequently,

$$Q(N) = \left(\frac{1}{4} + o(1)\right) N.$$

Proof. The proof has three steps. First we give a construction of size $\lfloor (N+2)/4 \rfloor$. Then we prove a uniform counting lemma for squarefree values of $x+y$ in a fixed residue class modulo 4. Finally we apply the lemma to the residue classes of A modulo 4.

Step 1: the lower bound

Let

$$R_2(N) = \{n \leq N : n \equiv 2 \pmod{4}\}.$$

If $a, b \in R_2(N)$ and $a \neq b$, then

$$a + b \equiv 2 + 2 \equiv 0 \pmod{4}.$$

Thus $4 \mid a + b$. Since $4 = 2^2$ is the square of a prime, $a + b$ is not squarefree. Therefore $R_2(N)$ is non-squarefree-sum, and hence

$$Q(N) \geq |R_2(N)| = \left\lfloor \frac{N+2}{4} \right\rfloor.$$

Step 2: a shifted squarefree-counting lemma

We need the following standard identity. Let μ denote the Möbius function. For every positive integer m ,

$$\mathbf{1}_{\text{sqfree}}(m) = \sum_{d^2 \mid m} \mu(d).$$

Indeed, this sum is 1 if no prime square divides m , and is 0 otherwise.

Lemma 1 (Squarefree values in one residue class). *Fix an integer y with $1 \leq y \leq N$ and fix a residue class $i \pmod{4}$. Suppose*

$$i + y \not\equiv 0 \pmod{4}.$$

Then, uniformly in i, y , and N ,

$$\#\{1 \leq x \leq N : x \equiv i \pmod{4}, x + y \text{ is squarefree}\} = \frac{2}{\pi^2}N + O(\sqrt{N}).$$

Proof. For $x \equiv i \pmod{4}$, the assumption $i + y \not\equiv 0 \pmod{4}$ gives

$$x + y \not\equiv 0 \pmod{4}.$$

Hence $4 \nmid x + y$. In particular, if d is even, then d^2 is divisible by 4, so $d^2 \nmid x + y$. Thus only odd d can contribute to the squarefree-indicator identity.

Since $1 \leq x \leq N$ and $1 \leq y \leq N$, we have $2 \leq x + y \leq 2N$. Therefore any d with $d^2 \mid x + y$ satisfies $d \leq \sqrt{2N}$. Thus

$$\begin{aligned} S_i(y; N) &:= \#\{1 \leq x \leq N : x \equiv i \pmod{4}, x + y \text{ is squarefree}\} \\ &= \sum_{\substack{1 \leq x \leq N \\ x \equiv i \pmod{4}}} \sum_{d^2 \mid x+y} \mu(d) \\ &= \sum_{\substack{d \leq \sqrt{2N} \\ d \text{ odd}}} \mu(d) \#\{1 \leq x \leq N : x \equiv i \pmod{4}, x \equiv -y \pmod{d^2}\}. \end{aligned}$$

Now let d be odd. Since $(4, d^2) = 1$, the two congruences

$$x \equiv i \pmod{4}, \quad x \equiv -y \pmod{d^2}$$

combine, by the Chinese remainder theorem, into one residue class modulo $4d^2$. The number of integers $1 \leq x \leq N$ in any fixed residue class modulo $4d^2$ is

$$\frac{N}{4d^2} + O(1),$$

with an absolute implied constant. Hence

$$\begin{aligned} S_i(y; N) &= \sum_{\substack{d \leq \sqrt{2N} \\ d \text{ odd}}} \mu(d) \left(\frac{N}{4d^2} + O(1) \right) \\ &= \frac{N}{4} \sum_{\substack{d \leq \sqrt{2N} \\ d \text{ odd}}} \frac{\mu(d)}{d^2} + O\left(\sum_{d \leq \sqrt{2N}} 1 \right) \\ &= \frac{N}{4} \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\mu(d)}{d^2} + O\left(N \sum_{d > \sqrt{2N}} \frac{1}{d^2} \right) + O(\sqrt{N}) \\ &= \frac{N}{4} \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\mu(d)}{d^2} + O(\sqrt{N}). \end{aligned}$$

The Euler product is

$$\begin{aligned} \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\mu(d)}{d^2} &= \prod_{p \geq 3} \left(1 - \frac{1}{p^2} \right) \\ &= \frac{\prod_p (1 - p^{-2})}{1 - 2^{-2}} \\ &= \frac{\zeta(2)^{-1}}{3/4} \\ &= \frac{6/\pi^2}{3/4} \\ &= \frac{8}{\pi^2}. \end{aligned}$$

Therefore

$$S_i(y; N) = \frac{N}{4} \cdot \frac{8}{\pi^2} + O(\sqrt{N}) = \frac{2}{\pi^2} N + O(\sqrt{N}),$$

as required. □

Step 3: residue-class structure of large examples

For $r \in \{0, 1, 2, 3\}$, define

$$A_r = \{a \in A : a \equiv r \pmod{4}\}, \quad a_r = |A_r|.$$

We shall use the lemma to show that any extremal set must, for all sufficiently large N , be contained entirely in one of the two even residue classes modulo 4.

Let

$$\eta = \frac{1}{4} - \frac{2}{\pi^2}.$$

Numerically, $\eta \approx 0.04736$.

Corollary 1 (A residue-class bound). *Suppose $A \subseteq [N]$ is non-squarefree-sum. If $A_j \neq \emptyset$ and*

$$i + j \not\equiv 0 \pmod{4},$$

then

$$a_i \leq \eta N + O(\sqrt{N}).$$

The implied constant is absolute.

Proof. Choose $y \in A_j$. Consider all $x \leq N$ with $x \equiv i \pmod{4}$. The total number of such x is

$$\frac{N}{4} + O(1).$$

By the lemma, the number of such x for which $x + y$ is squarefree is

$$\frac{2}{\pi^2}N + O(\sqrt{N}),$$

because $x + y \equiv i + j \not\equiv 0 \pmod{4}$.

Therefore the number of such x for which $x + y$ is not squarefree is

$$\frac{N}{4} - \frac{2}{\pi^2}N + O(\sqrt{N}) = \eta N + O(\sqrt{N}).$$

Now every $x \in A_i$ with $x \neq y$ must lie among these non-squarefree values, since A is non-squarefree-sum. If $i \neq j$, then $x \neq y$ automatically. If $i = j$, the single possible element $x = y$ is not constrained by the distinct-pair condition, and contributes at most one extra element. This extra 1 is absorbed into the $O(\sqrt{N})$ term. Hence

$$a_i \leq \eta N + O(\sqrt{N}).$$

□

We now classify the possible residue supports of A .

If A is contained in A_0 , then

$$|A| \leq \#\{n \leq N : n \equiv 0 \pmod{4}\} = \left\lfloor \frac{N}{4} \right\rfloor.$$

If A is contained in A_2 , then

$$|A| \leq \#\{n \leq N : n \equiv 2 \pmod{4}\} = \left\lfloor \frac{N+2}{4} \right\rfloor.$$

These are the two exceptional cases, and both are compatible with the desired upper bound.

It remains to show that no other residue support can be extremal for large N .

First suppose that A contains an element in an odd residue class. If $A_1 \neq \emptyset$, then applying the corollary with $(i, j) = (1, 1)$ gives $a_1 \leq \eta N + O(\sqrt{N})$, because $1 + 1 \equiv 2 \not\equiv 0 \pmod{4}$. Similarly, if $A_3 \neq \emptyset$, then applying the corollary with $(i, j) = (3, 3)$ gives $a_3 \leq \eta N + O(\sqrt{N})$.

Moreover, if an even class A_i with $i \in \{0, 2\}$ is nonempty and some odd class A_j with $j \in \{1, 3\}$ is nonempty, then $i + j$ is odd, hence not congruent to 0 (mod 4). The corollary gives

$$a_i \leq \eta N + O(\sqrt{N}).$$

Thus, if any odd residue class occurs, every nonempty residue class has size at most $\eta N + O(\sqrt{N})$.

Second suppose that no odd residue class occurs. If A is not contained in just A_0 or just A_2 , then both A_0 and A_2 are nonempty. Applying the corollary with $(i, j) = (0, 2)$ and with $(i, j) = (2, 0)$ gives

$$a_0, a_2 \leq \eta N + O(\sqrt{N}),$$

since $0 + 2 \equiv 2 \not\equiv 0 \pmod{4}$.

Consequently, unless A is contained entirely in A_0 or entirely in A_2 , we have

$$|A| = a_0 + a_1 + a_2 + a_3 \leq 4\eta N + O(\sqrt{N}).$$

But

$$4\eta = 4 \left(\frac{1}{4} - \frac{2}{\pi^2} \right) = 1 - \frac{8}{\pi^2} \approx 0.18943 < \frac{1}{4}.$$

Therefore, for all sufficiently large N ,

$$4\eta N + O(\sqrt{N}) < \left\lfloor \frac{N+2}{4} \right\rfloor.$$

By Step 1, there exists a non-squarefree-sum set of size $\lfloor (N+2)/4 \rfloor$, namely $R_2(N)$. Hence, for all sufficiently large N , an extremal set cannot fall into the non-exceptional case. It must be contained entirely in A_0 or entirely in A_2 .

Finally, among these two residue classes,

$$\#\{n \leq N : n \equiv 0 \pmod{4}\} = \left\lfloor \frac{N}{4} \right\rfloor$$

and

$$\#\{n \leq N : n \equiv 2 \pmod{4}\} = \left\lfloor \frac{N+2}{4} \right\rfloor.$$

Since

$$\left\lfloor \frac{N+2}{4} \right\rfloor \geq \left\lfloor \frac{N}{4} \right\rfloor,$$

the largest possible size is

$$Q(N) = \left\lfloor \frac{N+2}{4} \right\rfloor$$

for all sufficiently large N . □

Remark 1 (What the proof uses). *The proof is insensitive to the diagonal sums $a + a$. The only place where the distinction between pair-sums with $a \neq b$ and all sums including $a = a$ enters is in the corollary, where a single possible element $x = y$ is allowed and absorbed into the error term. Thus the argument directly addresses the distinct-pair formulation of Problem 4.*

References

- [1] Przemek Chojacki and GPT-5.5 Pro, *Ten New Erdős-style problems in number theory and combinatorics*, curated problem list, May 2026.